

Calculus 2 - Homework 1 (Worked Solutions)

1.

The denominator $x^2 - 4$ factors (difference of squares), so this is partial fractions, NOT an arctan/inverse-trig form (that would need $x^2 + 4$). Write $1/(x^2 - 4) = 1/((x-2)(x+2)) = A/(x-2) + B/(x+2)$.

$1 = A(x+2) + B(x-2)$. $x = 2$: $1 = 4A \rightarrow A = 1/4$. $x = -2$: $1 = -4B \rightarrow B = -1/4$.

$\int [(1/4)/(x-2) - (1/4)/(x+2)] dx = (1/4) \ln |(x-2)/(x+2)| + C$.

ANSWER: $(1/4) \ln |(x-2)/(x+2)| + C$

2.

$a_n = (n+1)/(n^3 + 2)$. Ratio:

$a_{n+1}/a_n = [(n+2)/((n+1)^3 + 2)] * [(n^3 + 2)/(n+1)]$.

As $n \rightarrow \infty$ this ratio $\rightarrow 1$ (numerator and denominator are both cubic in n after expanding), so $L = 1$ and the Ratio Test is INCONCLUSIVE.

Switch to Limit Comparison with $b_n = 1/n^2$ (since $a_n \sim n/n^3 = 1/n^2$):

$\lim [(n+1)/(n^3+2)] / [1/n^2] = \lim n^2(n+1)/(n^3+2) = 1$ (finite, positive).

$\sum 1/n^2$ converges ($p = 2$), so $\sum (n+1)/(n^3 + 2)$ converges.

ANSWER: converges (ratio test inconclusive with $L = 1$; limit comparison with $\sum 1/n^2$)

3.

The $\sqrt{9 - x^2}$ form is the classic trig-substitution signal, but the lone factor x in the numerator makes ordinary substitution work directly, so no trig substitution is needed.

Let $u = 9 - x^2$, so $du = -2x dx$, i.e. $x dx = -(1/2) du$.

The integral becomes $\int -(1/2) u^{-1/2} du = -(1/2) * 2 u^{1/2} + C$

$= -u^{1/2} + C = -\sqrt{u} + C$.

Back-substitute $u = 9 - x^2$: $-\sqrt{9 - x^2} + C$.

Check: $d/dx[-\sqrt{9-x^2}] = -(1/(2\sqrt{9-x^2}))(-2x) = x/\sqrt{9-x^2}$.

ANSWER: $-\sqrt{9 - x^2} + C$

4.

For large n the terms behave like $n/n^3 = 1/n^2$, so compare with $b_n = 1/n^2$.

Limit Comparison: $\lim_{n \rightarrow \infty} [n/(n^3+1)] / [1/n^2]$

$= \lim n * n^2 / (n^3 + 1) = \lim n^3/(n^3 + 1) = 1$, a finite positive number.

Since $\sum 1/n^2$ converges ($p = 2$), the Limit Comparison Test gives that $\sum n/(n^3 + 1)$ converges too.

ANSWER: converges (limit comparison with $\sum 1/n^2$, ratio 1)

5.

The product of a trig function and an exponential can tempt an integration-by-parts attack, but that leads in circles here. Instead see that the exponent's derivative sits outside: $d/dx(\sin x) = \cos x$.

Let $u = \sin x$, so $du = \cos x dx$.

The integral becomes $\int e^u du = e^u + C$.

Back-substitute $u = \sin x$: $e^{(\sin x)} + C$.

Check: $d/dx[e^{(\sin x)}] = e^{(\sin x)} \cos x$, the integrand.

ANSWER: $e^{(\sin x)} + C$

6.

Form $\sqrt{x^2 + a^2}$ with $a = 1$. Substitute $x = \tan \theta$,

$dx = \sec^2 \theta d\theta$, $\sqrt{x^2 + 1} = \sec \theta$.

$\int \sec \theta * \sec^2 \theta d\theta = \int \sec^3 \theta d\theta$.

Using the standard reduction, $\int \sec^3 \theta \, d\theta = \frac{1}{2}(\sec \theta \tan \theta + \ln|\sec \theta + \tan \theta|)$.
 Back-substitute: $\tan \theta = x$, $\sec \theta = \sqrt{x^2 + 1}$.
 Result = $\frac{1}{2}(x \sqrt{x^2 + 1} + \ln|\sqrt{x^2 + 1} + x|) + C$.
 ANSWER: $\frac{x}{2} \sqrt{x^2 + 1} + \frac{1}{2} \ln(x + \sqrt{x^2 + 1}) + C$

7.

A student may try $u = x^2 - 9$, but $du = 2x \, dx$ and there is no x factor in the numerator, so substitution stalls. The $\sqrt{x^2 - a^2}$ form calls for a secant trig substitution.

Let $x = 3 \sec \theta$, $dx = 3 \sec \theta \tan \theta \, d\theta$, and
 $\sqrt{x^2 - 9} = \sqrt{9 \sec^2 \theta - 9} = 3 \tan \theta$ (θ in $[0, \pi/2)$).
 $\int \frac{3 \sec \theta \tan \theta}{3 \tan \theta} \, d\theta = \int \sec \theta \, d\theta = \ln|\sec \theta + \tan \theta| + C$.

With $\sec \theta = x/3$ and $\tan \theta = \sqrt{x^2 - 9}/3$:
 $= \ln|x/3 + \sqrt{x^2 - 9}/3| + C = \ln|x + \sqrt{x^2 - 9}| + C'$
 (the constant $\ln 3$ is absorbed).
 ANSWER: $\ln|x + \sqrt{x^2 - 9}| + C$

8.

Parts applied twice. First $u = x^2$, $dv = e^x \, dx$, so $du = 2x \, dx$, $v = e^x$.
 $\int x^2 e^x \, dx = x^2 e^x - \int 2x e^x \, dx$.
 For $\int 2x e^x \, dx$ apply parts again with $u = 2x$, $dv = e^x \, dx$:
 $\int 2x e^x \, dx = 2x e^x - \int 2 e^x \, dx = 2x e^x - 2 e^x$.
 Combine: $x^2 e^x - (2x e^x - 2 e^x) + C = (x^2 - 2x + 2) e^x + C$.
 Check: $d/dx[(x^2 - 2x + 2)e^x] = (2x - 2)e^x + (x^2 - 2x + 2)e^x = x^2 e^x$.
 ANSWER: $(x^2 - 2x + 2) e^x + C$

9.

Form $\sqrt{a^2 - x^2}$ with $a = 3$. Substitute $x = 3 \sin(\theta)$,
 $dx = 3 \cos(\theta) \, d\theta$, and $\sqrt{9 - x^2} = \sqrt{9 - 9 \sin^2 \theta} = 3 \cos(\theta)$.
 $\int \frac{1}{3 \cos \theta} \cdot 3 \cos \theta \, d\theta = \int 1 \, d\theta = \theta + C$.
 Since $x = 3 \sin \theta$, $\theta = \arcsin(x/3)$.
 ANSWER: $\arcsin(x/3) + C$

10.

Compare with the harmonic series $b_n = 1/n$.
 Limit Comparison: $\lim_{n \rightarrow \infty} \frac{1/(2n+1)}{1/n} = \lim_{n \rightarrow \infty} \frac{n}{2n+1} = 1/2$, a finite positive number.
 The harmonic series $\sum 1/n$ diverges, so by the Limit Comparison Test $\sum 1/(2n+1)$ diverges as well.
 ANSWER: diverges (limit comparison with harmonic sum $1/n$, ratio $1/2$)